



Exercise 1.

Perform the following calculations:

a) $(8 - 3 \times 2)^2$

b) $8 - 3 \times 2^2$

c) $(8 - 3) \times 2^2$

d) $8 - (3 \times 2)^2$

e) $2 + 4 \times 7^2$

f) $(-3)^2 + 2 \times 5^2$

g) $-3(-8 + 6)^5$

h) $[3 - 2(-4)]^2 \times 3$

i) $(4 + 2)^2 \div 9$

j) $16 \div (9 - 7)^2$

k) $\frac{54}{5 \times 9 - 6^2}$

l) $\frac{(5^2 - 3 \times 7)^2}{10 - 2^3}$

Exercise 2.

Write True or False (with justification):

1) $36^3 = 6^6$

2) $27^4 = 3^{12}$

3) $2^{30} > 3^{20}$

4) $8^{24} < 16^{18}$

5) $125^6 - 25^9 = 0$

6) $2^6 + 2^6 = 2^7$

Exercise 3.

Find the **sign** of each of the following expressions without doing any calculations.

$$A = (-7)^3 \times (-2^4)$$

$$B = \frac{-5^6}{(-9)^9}$$

$$C = \frac{(-5)^2 \times (-2)^3}{(-1)^6 \times (-7)^9}$$

$$D = [-(-3)^2]^7 \times [(-5)^4]^5$$

$$E = \frac{(-3)^5 \times (-4)^2}{(-1)^3 \times (-2)^7}$$

Exercise 4.

Calculate (rapidly):

1) $(0.25)^5 \times 4^5$

2) $\left(\frac{1}{4}\right)^3 \times 12^3$

3) $\frac{(-3)^5 \times (-4)^2}{(-1)^3 \times (-2)^7}$

4) $\left(\frac{7}{2}\right)^3 \div \left(\frac{21}{4}\right)^3$

5) $\frac{2^4 \times 3^2 \times 5^4}{4^2 \times 25^2 \times 3}$

6) $5^6 \times 3^2 \times 2^6$

Exercise 5.

Write in the form of a single power:

1) $(-3)^6 \times (3)^3 \times (-3)^2$

2) $\frac{(-5)^7}{(-5)^3}$

3) $\frac{5 \times 5^2 \times 5^4}{5^3 \times 5^0}$

4) $\frac{(a^8)^2 \times (-a)^4}{[(-a)^3]^4}$

5) $(0.25)^4 \times 4^6$

Exercise 6.

Write in the form of a power of 10:

1) 2×500

3) $\frac{1}{0.0001}$

5) $\frac{1}{625 \times 1.6}$

2) 0.4×2500

4) 80×0.125

6) $\frac{1}{4} \times \frac{1}{25}$

Exercise 7.

Simplify:

a) $\frac{3 \times 10^3 \times 0.15}{15 \times 10^{-1} \times 2}$

b) $\frac{4.5 \times 0.21 \times 10^3}{0.5 \times 70}$

c) $\frac{(-0.2)^4 \times (0.5)^3 \times (0.8)^2 \times 10^5}{0.04^2 \times (-2)^3 \times (-5)^2}$

d) $\frac{2.25^3 \times 4.9^4 \times 0.8}{0.14^3 \times 10.5^5}$

e) $\left(-\frac{5}{2}\right)^2 \times \left(\frac{3}{5}\right)^2 \times (6)$

f) $\left(\frac{1}{3} - \frac{2}{6}\right)^7 \times \left(\frac{3}{2} - \frac{8}{4}\right)$

Exercise 8.

Calculate then write in scientific notation:

a) $\frac{3 \times 10^5 \times 40 \times 10^{-6}}{8 \times 10^{-8}}$; b) $6.5 \times 10^{-6} \times 10^{24} \times 4$

Exercise 9.

In our blood we have red blood cells, white blood cells, and platelets.

Mrs. Lamia is in good health, she has:

- 5 million of red blood cells per mm^3 of blood
- 7 million of white blood cells per mm^3 of blood
- 275000 platelets per mm^3 of blood

1) Write each of the above numbers in scientific notation.

2) a) Complete the following:

$$1\text{L} = \dots \text{dm}^3 = \dots \text{cm}^3 = \dots \text{mm}^3$$

b) Calculate the number of platelets contained in 5 L of Mrs. Lamia's blood.